

Linear Search Problems (35 Different Constraint Environments)

1. Basic Linear Search

Problem: Given an array of integers and a target value, find the index of the target using linear search. If not found, return -1.

Constraints: $1 \leq N \leq 100$

Sample Input:

$N = 5$

Array = [4, 2, 7, 1, 9]

Target = 7

Sample Output:

2

2. Search with Count

Problem: Find how many times a given integer appears in an array using linear search.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [1, 2, 3, 2, 2, 5]

Target = 2

Sample Output:

3

3. First Occurrence

Problem: Find the first occurrence index of a target value.

Constraints: $1 \leq N \leq 10^4$

Sample Input:

Array = [5, 6, 5, 6, 7]

Target = 5

Sample Output:

0

4. Last Occurrence

Problem: Find the last occurrence index of a target value.

Constraints: $1 \leq N \leq 10^4$

Sample Input:

Array = [5, 6, 5, 6, 7]

Target = 6

Sample Output:

3

5. Search in Reverse Order

Problem: Given an array where the **last element is considered to be stored at index 0**, the second last at index 1, and so on, search for a given target element by **traversing the array from the end**.

If the element is found, return its corresponding index based on this reverse indexing system; otherwise, return **-1**.

Constraints: $1 \leq N \leq 500$

Sample Input:

Array = [10, 20, 30, 40]

Target = 20

Sample Output: 2

6. Minimum Value Search

Problem: Find the minimum value in an array using linear search.

Constraints: $1 \leq N \leq 10^5$

Sample Input:

Array = [8, 3, 6, 2, 9]

Sample Output:

2

7. Maximum Value Search

Problem: Find the maximum value in an array using linear search.

Constraints: $1 \leq N \leq 10^5$

Sample Input:

Array = [8, 3, 6, 2, 9]

Sample Output:

9

8. Even Number Search

Problem: Find the first even number in an array.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [3, 7, 5, 8, 9]

Sample Output:

8

9. Odd Number Count

Problem: Count the total odd numbers in an array.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [3, 7, 5, 8, 9]

Sample Output:

4

10. Search with Break Condition

Problem: Stop searching once the target is found and return its position.

Constraints: $1 \leq N \leq 100$

Sample Input:

Array = [11, 22, 33, 44]

Target = 33

Sample Output:

2

11. Search in Sorted Array (Linear)

Problem: Search an element in a sorted array using linear search.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [1, 3, 5, 7, 9]

Target = 7

Sample Output:

3

12. Index Sum Search

Problem: Find the sum of indices where a target appears.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [2, 4, 2, 6, 2]

Target = 2

Sample Output:

6

13. Search with Limited Range

Problem: Search a target only between given index range.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [5, 10, 15, 20, 25]

Target = 20

Range = 1 to 3

Sample Output:

3

14. Search Negative Number

Problem: Find the first negative number in an array.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [4, 6, -2, 8]

Sample Output: -2

15. Frequency Check

Problem: Check if a number appears more than once using linear search.

Constraints: $1 \leq N \leq 1000$

Sample Input:

Array = [1, 2, 3, 2, 4]

Target = 2

Sample Output:

Yes

16. Second Largest Element Index

Problem:

Find the index of the **second largest distinct element** in the array using linear search. If it does not exist, return -1.

Constraints:

$1 \leq N \leq 10^5$

Sample Input:

Array = [4, 7, 2, 7, 5]

Sample Output: 4

17. Closest Element to Target

Problem:

Find the element whose value is **closest to a given target**. If multiple elements are equally close, return the one with the **smallest index**.

Constraints:

$1 \leq N \leq 10^5$

Sample Input:

Array = [1, 4, 6, 8]

Target = 5

Sample Output:

4

18. Majority Element Check

Problem:

Check whether an element appears **more than N/2 times** in the array using linear search. Print the element if it exists, otherwise print -1.

Constraints:

$$1 \leq N \leq 10^4$$

Sample Input:

Array = [3, 3, 4, 3, 3]

Sample Output:

3

19. First Repeating Element Index**Problem:**

Find the index of the **first element that repeats later** in the array. If no such element exists, return -1.

Constraints:

$$1 \leq N \leq 10^4$$

Sample Input:

Array = [10, 5, 3, 4, 3, 5, 6]

Sample Output:

1

20. Smallest Missing Positive Integer**Problem:**

Find the **smallest positive integer missing** from the array using linear search.

Constraints:

$$1 \leq N \leq 10^4$$

Sample Input:

Array = [3, 4, -1, 1]

Sample Output:

2

21. Peak Element Search

Problem:

Find the index of a **peak element** (greater than its immediate neighbors). Assume the first and last elements have only one neighbor.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [1, 3, 20, 4, 1, 0]

Sample Output:

2

22. Maximum Difference Pair**Problem:**

Find the **maximum difference** $\text{Array}[j] - \text{Array}[i]$ such that $j > i$ using linear search.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [2, 3, 10, 6, 4, 8, 1]

Sample Output:

8

23. Longest Consecutive Sequence Length**Problem:**

Find the length of the **longest sequence of consecutive integers** present in the array using linear search.

Constraints:

$$1 \leq N \leq 10^4$$

Sample Input:

Array = [1, 9, 3, 10, 4, 20, 2]

Sample Output:

4

24. First Element Greater Than All Left Elements

Problem:

Find the index of the **first element that is greater than all elements to its left**.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [2, 3, 5, 1, 7, 6]

Sample Output:

4

25. Minimum Distance Between Two Targets**Problem:**

Given two target values X and Y, find the **minimum index distance** between their occurrences in the array. If either is missing, return -1.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [3, 5, 4, 2, 6, 5, 6, 4, 8]

X = 5, Y = 6

Sample Output:

1

26. Index of Maximum Sum Prefix**Problem:**

Find the index up to which the **prefix sum is maximum**.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [2, -1, 3, -2, 4]

Sample Output:

4

27. First Element with Exactly K Occurrences

Problem:

Find the **first element** that appears **exactly K times** in the array. If none exists, return -1.

Constraints:

$$1 \leq N \leq 10^4$$

Sample Input:

Array = [1, 2, 2, 3, 3, 3]

K = 2

Sample Output:

2

28. Count Elements Smaller Than All Right Elements**Problem:**

Count how many elements are **strictly smaller than every element to their right**.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [5, 1, 4, 3, 6]

Sample Output:

2

29. First Element Whose Frequency Is Prime**Problem:**

Find the **first element** in the array whose **frequency is a prime number**. If none exists, return -1.

Constraints:

$$1 \leq N \leq 10^4$$

Sample Input:

Array = [2, 3, 2, 3, 3, 4]

Sample Output:

2

30. Longest Increasing Prefix Length

Problem:

Find the length of the **longest strictly increasing prefix** of the array.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [1, 2, 3, 2, 5]

Sample Output:

3

31. First Element Greater Than Average**Problem:**

Find the index of the **first element greater than the average** of all array elements.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [2, 4, 6, 8]

Sample Output:

2

32. Element with Minimum Absolute Difference**Problem:**

Find the element that has the **minimum absolute difference** with a given target.

If multiple elements qualify, return the one with the **largest value**.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [1, 3, 5, 7]

Target = 6

Sample Output:

5

33. Count Elements Equal to Product of Neighbors

Problem:

Count how many elements are equal to the **product of their immediate neighbors**.

Constraints:

$$3 \leq N \leq 10^5$$

Sample Input:

Array = [2, 3, 6, 2]

Sample Output:

1

34. Index of First Unsorted Element**Problem:**

Find the index of the **first element that breaks ascending order**.

If the array is sorted, return -1.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [1, 2, 3, 5, 4, 6]

Sample Output:

4

35. Maximum Frequency Element (Earliest Index)**Problem:**

Find the element with the **highest frequency**.

If multiple elements have the same maximum frequency, return the one with the **smallest index**.

Constraints:

$$1 \leq N \leq 10^5$$

Sample Input:

Array = [4, 1, 2, 2, 3, 1, 1]

Sample Output:

1